



**NANYANG
TECHNOLOGICAL
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SINGAPORE

NTU Academy for Professional
and Continuing Education

UI-UX Principles

Module 6

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UI and UX first principles

The design foundations for the website you are about to build

A good-looking screen does not guarantee a good experience

A café's website has beautiful photographs, an elegant menu, and clearly listed opening hours.

The hours are from last year. You travel there on a Monday and find the café closed.

The screen was excellent, and the experience still failed.

Two terms: UI (user interface) and UX (user experience)

The user interface is what the product looks like and how it behaves on the screen

The user experience is everything around it: finding the site, trusting it, and what happens away from the screen

A product can have an excellent interface and still deliver a poor experience, as the café example shows

Source: The term "user experience" was coined by Don Norman at Apple in the mid-1990s

The plan for the next 45 minutes

After these four parts, you will build and publish a five-page café website.

1. Five questions you can ask of any design
2. How a website differs from an app
3. How a phone differs from a laptop
4. Writing design instructions for an AI (artificial intelligence) assistant

FIVE QUESTIONS · 1 OF 4

Five questions to ask any design

What do I see in the first three seconds?

Readers scan a page instead of reading it — eye-tracking studies show an F-shaped scanning path

Size, position, and contrast decide what the eye notices first

The most important element on the page should also be the most prominent one

When every element is emphasised, the reader cannot tell which one matters

Source: Nielsen Norman Group eye-tracking research, 2006

What is this page for?

Every page should have one job that you can state in one sentence

The page's main button or link — its call to action — should match that job

Style the remaining links so they are clearly less prominent than the main action

Example: the job of a café's home page is to convince a visitor to walk in.

Where does this surprise the user?

"Users spend most of their time on other sites. They prefer your site to work the same way as all the other sites they already know."

Source: Jakob Nielsen, 2000

What can be removed?

The time a person needs to decide grows with the number of choices in front of them

A home page with **one** clear call to action outperforms one with seven equally weighted links

Keep the main choices visible, group related items, and move rarely used options behind a "more" link

Source: Hick's Law – Hick and Hyman, 1952

Can a thumb hit this?

The time needed to reach a button depends on how far away it is and how big it is

Make touch targets **44 by 44 pixels (px)**; the WCAG (Web Content Accessibility Guidelines) minimum is 24 px

Targets at the edge of the screen are the easiest to hit, because a finger or cursor cannot overshoot them

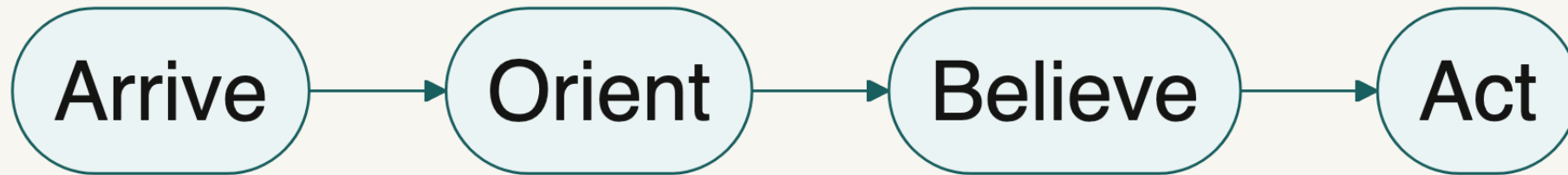
Source: Fitts's Law – Paul Fitts, 1954

The five questions together

Question	Principle
What do I see in three seconds?	Visual hierarchy
What is this page for?	One job per page
Where does this surprise the user?	Jakob's Law
What can be removed?	Hick's Law
Can a thumb hit this?	Fitts's Law

How a website differs from an app

A website serves a visitor



A first-time visitor is deciding whether to trust the business, so clear content order and believable details do the work.

An app serves a returning user

The user has already chosen the product and opens the app to complete a task

The same task repeats until it becomes a habit

Speed, remembered settings, and consistent screens matter most

Anything unfamiliar slows the user down on every single visit

The same questions, weighted differently

	Website	App
Who arrives	A first-time visitor, deciding	A returning user, doing
The page's job	Persuade and inform	Finish the task quickly
What matters most	Clarity and trust	Speed and habit

The café site you are about to build is a website: its goal is to convince a visitor, so clarity and trust come first.

How a phone differs from a laptop

Designing for a thumb on a small screen

A mouse pointer is precise; a thumb covers about a centimetre of glass and hides what it taps

Make every button and link at least 44 pixels (px) tall, and avoid features that require hovering with a mouse

A phone shows one narrow column, so side-by-side layouts stack vertically

Phone users are often standing, holding the phone in one hand, with little patience

Design the phone view first

Start the design at a phone width of **375 pixels (px)**, before the laptop width of 1280 px

The small screen forces you to rank every element by importance

The elements you keep in the phone view are the ones that matter most

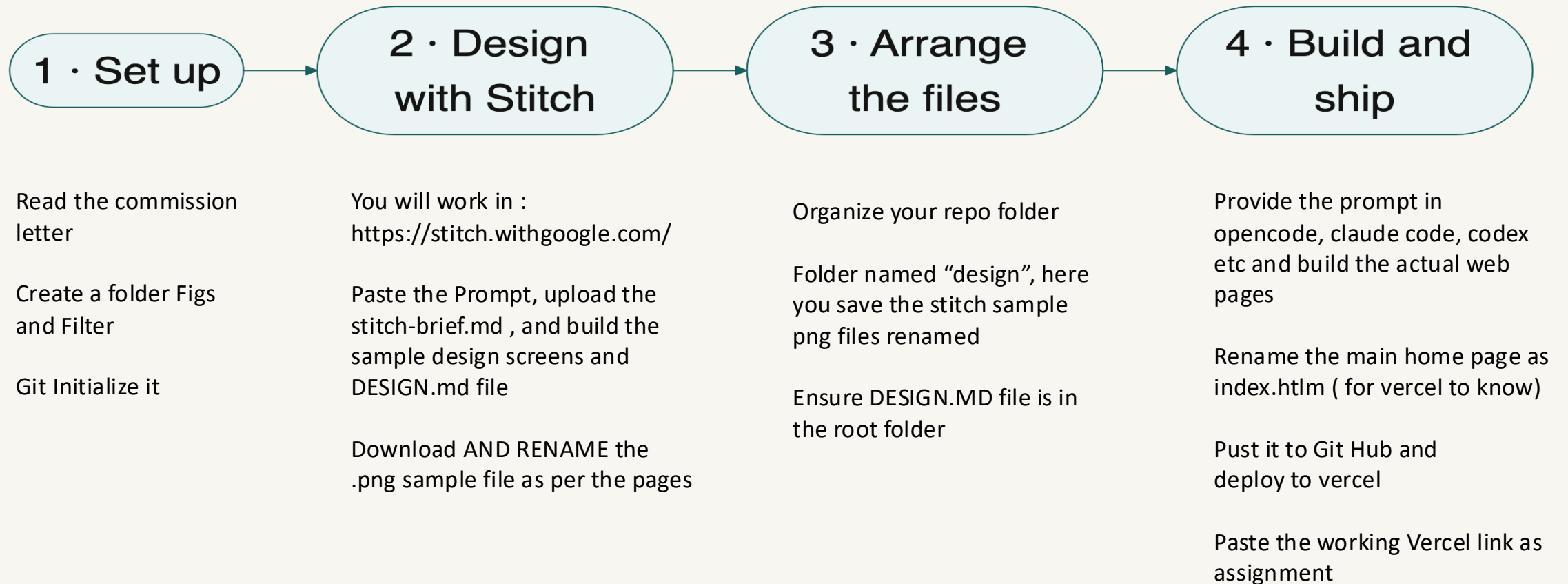
Widening the layout for a laptop is then straightforward, because the ranking is already done

Turning principles into instructions



The route from pack to live site

The next four slides take one phase each.



Phase 1 — set up the project

Create a project folder on your computer — it will become your GitHub repository

Copy the `content` folder from your pack into it: the copy, the brand file, and the photographs

Phase 2 — design with Google Stitch

Upload the design brief and paste the master prompt; Stitch generates all five screens

Judge every screen with the five questions; ask for one testable change at a time

Download the export — a zip with one folder per screen, plus DESIGN.md

Phase 3 — arrange the design files

Save DESIGN.md in the project root, keeping its name

Create a folder called `design` and move the five screen images into it

Rename each image to match the page it becomes: index.png, menu.png, catering.png, about.png, visit.png



Phase 4 — build, ship, and check

Point your AI (artificial intelligence) app at the project folder and paste the build prompt

The AI writes five HTML (HyperText Markup Language) pages, one stylesheet, and an images folder

Quote any failed rule back and ask for only that fix

Push to GitHub, deploy on Vercel, and run the checklist on a phone and a laptop

What a good DESIGN.md contains

The audience for the site, and the one job of each page

The order of importance on each page: what comes first, what is biggest, and the single main action

How the layout behaves at each screen width: what collapses, what stacks, what stays

The exact colours, fonts, and tone of voice, so the AI (artificial intelligence) assistant does not invent its own

Write instructions the AI can be tested against

Vague instruction	Testable instruction
"Make it clean and modern"	"Give each page one primary call to action"
"It should work on mobile"	"Collapse the navigation below 768 pixels (px); no sideways scrolling at 375 px"
"Easy to read"	"Set body text at 16 px or larger"

You can check every instruction on the right with your own eyes on the finished site.

Live demonstration — one Google Stitch layout

- We give Google Stitch one prompt, and it returns a page layout
- Together we judge the layout using the five questions
- We list the changes we would ask for before building it

Your turn: the Fig & Filter website

- Your pack: the client's commission letter, the build handbook, the content folder, and the route card
- The path: Google Stitch layouts, then your DESIGN.md, then the AI (artificial intelligence) build, then GitHub, then Vercel
- The site is complete when every page passes the five questions on both a phone and a laptop

Version Management

Month/Year	Version	Changes
May 2026	0.0	New Module Design
July 2026	1.0	Updated content and case study